Python Data Analysis Notes

1. Python Libraries (Page 1)

- **pandas** → Data manipulation (DataFrames)
- **numpy** → Numerical operations
- **matplotlib** → Basic visualization
- **seaborn** → Advanced statistical visualization
- **scikit-learn** → Machine learning
- **plotly** → Interactive dashboards
- **requests** → API data fetching

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2. Data Structures

- **List** → Mutable, ordered
- **Tuple** → Immutable, ordered
- **Dictionary** → Key-value pairs (fast lookup)

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3. Handling Missing Data (Page 2)

- Check → df.isnull().sum()
- Fill:
  - Mean / Median
  - Forward fill (ffill)
  - Backward fill (bfill)
- Remove → df.dropna()

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4. Reading Data

- Load → pd.read_csv()
- View → df.head()
- Info → df.info()
- Stats → df.describe()

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5. loc vs iloc

- **loc[]** → Label-based
- **iloc[]** → Index-based

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6. Filtering Data (Page 3)

- Single condition:
- df[df['Amount'] > 500]
- Multiple conditions:
- df[(cond1) & (cond2)]

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7. groupby()

- Used for aggregation:
  - Sum
  - Mean
  - Count
    Example:

df.groupby('Region')['Sales'].sum()

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8. Merging Data

- merge() → SQL-like join
- Types:
  - Inner
  - Left
  - Right
  - Outer

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9. Lambda Functions

- One-line functions

lambda x: x * 0.9

- Used with apply()

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10. Data Visualization (Page 5)

- Line plot:

```
plt.plot(x, y)
```

- Used for trends (sales over time)

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11. apply() Function

- Apply custom function to column

```
df['col'].apply(lambda x: x*2)
```

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12. Pivot Table

- Summarizes data

```
df.pivot_table(values='Sales', index='Month', columns='Region')
```

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13. Data Formats (Page 7)

- **Wide** → Multiple columns
- **Long** → Single column with values
- Convert:
    - melt() → Wide → Long
    - pivot() → Long → Wide

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14. Correlation

- Formula:

```
df['A'].corr(df['B'])
```

- Values:
    - +1 → Strong positive
    - 0 → No relation
    - -1 → Strong negative

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15. Datetime Handling

- Convert:

```
pd.to_datetime()
```

- Extract:

- Year
- Month
- Day

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16. Handling Outliers (Page 9)

- Methods:
    - IQR
    - Z-score
    - Capping
- Visualization:
    - Boxplot
    - Histogram

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17. merge vs concat vs join

- **merge()** → key-based
- **concat()** → stack data
- **join()** → index-based

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18. Automation

- Use loops + functions
- Automate:
    - Cleaning
    - Processing multiple files

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19. Seaborn vs Matplotlib (Page 11)

- **Seaborn** → Better visuals
- **Matplotlib** → More control

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20. Handling Invalid Data

- Replace negatives
- Fill missing values

- Drop bad rows

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21. Key Analysis Tasks

- Top category → groupby + idxmax
- Monthly trends → groupby + plot
- Merge files → concat
- Dashboard → Plotly/Dash